

Simple Machines — Lesson 1

ICSE Class 6 Physics · 10 MCQs · 1 mark each · Name: _____ Date: _____

Syllabus: Work & energy, machines, principle of a machine, efficiency, ideal & actual machine, functions of a machine, mechanical advantage, levers, principle of a lever, orders of levers.

Instructions: Each question has one correct option. Tick or circle the letter of the best answer. Attempt all questions.

1. Work is said to be done only when:

- | | |
|---|-------------------------------|
| A. a body moves in the direction of the applied force | B. a body stays at rest |
| C. a force is applied to a body | D. energy is stored in a body |

2. The SI unit of work is the:

- | | |
|----------|-----------|
| A. watt | B. newton |
| C. joule | D. metre |

3. The capacity of a body to do work is called its:

- | | |
|---------------|-----------|
| A. power | B. energy |
| C. efficiency | D. force |

4. A machine used to lift a heavy load with a small effort is acting as a:

- | | |
|----------------------|---------------------|
| A. speed multiplier | B. energy source |
| C. direction changer | D. force multiplier |

5. Mechanical advantage is defined as the ratio:

- | | |
|-----------------------|------------------|
| A. Work out / Work in | B. Load x Effort |
| C. Load / Effort | D. Effort / Load |

6. An effort of 50 N lifts a load of 250 N using a machine. The mechanical advantage is:

- | | |
|--------|--------|
| A. 5 | B. 300 |
| C. 0.2 | D. 200 |

7. The efficiency of a machine is the ratio of:

- | | |
|-------------------|-------------------------------------|
| A. effort to load | B. useful work output to work input |
| C. VR to MA | D. load-arm to effort-arm |

8. An ideal machine is one whose efficiency is:

- | | |
|-------------------|-------------------|
| A. more than 100% | B. exactly 100% |
| C. 50% | D. less than 100% |

9. The fixed point about which a lever turns is the:

- | | |
|-----------|------------|
| A. load | B. arm |
| C. effort | D. fulcrum |

10. On a lever a load of 60 N acts 10 cm from the fulcrum and the effort-arm is 30 cm. The ideal effort is:

- | | |
|---------|----------|
| A. 20 N | B. 600 N |
| C. 6 N | D. 180 N |

Answer Key

1. A

2. C

3. B

4. D

5. C

6. A

7. B

8. B

9. D

10. A